SOLAR▸SWARM

Integration Guide
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CRM Integration Guide

Overview: crm_router Abstraction

File: `integrations/crm_router.py`

All agent code calls `crm_router` functions directly. The router selects the active CRM at runtime based on which credentials are configured.

Priority order: GoHighLevel → HubSpot → Salesforce

If GHL_API_KEY is set, all calls go to GHL. If only HubSpot is configured, calls go to HubSpot. Agents never need to know which platform is live.

```
from integrations.crm_router import crm_router

# These calls work identically regardless of which CRM is active
crm_router.get_contact("abc123")
crm_router.create_contact({"name": "Jane Smith", "email": "jane@example.com"})
crm_router.move_pipeline_stage("abc123", "qualified")
```

GoHighLevel (GHL) Setup

GHL is the primary CRM. All clients are expected to be on GHL.

Step 1: Get your API key

1. Log in to your GHL sub-account (not the agency account)
2. Go to **Settings → Integrations → API**
3. Copy the **Private Integration API Key**
4. Add to `.env`: `GHL_API_KEY=your-key-here`

Step 2: Get your Location ID

1. In GHL, go to **Settings → Business Profile**
2. The Location ID is visible in the URL: `app.gohighlevel.com/location/{LOCATION_ID}/...`
3. Add to `.env`: `GHL_LOCATION_ID=your-location-id`

Step 3: Get your Pipeline ID (for stage sync)

1. Go to **CRM → Pipelines**
2. Open the pipeline you want to track
3. The pipeline ID is in the URL
4. Add to `.env`: `GHL_PIPELINE_ID=your-pipeline-id`

Step 4: Configure Webhooks

GHL needs to be configured to send events to the Solar Swarm webhook server.

1. Go to **Settings → Integrations → Webhooks**

2. Click **Add Webhook**
3. Set the URL to your server: `https://your-server.com/webhook/new-lead`
4. Select the following events:

- **Contact Created** → maps to `/webhook/new-lead`
- **Opportunity Stage Change** → maps to `/webhook/stage-change`

1. Copy the **Signing Secret** and add to `.env`: `GHL_WEBHOOK_SECRET=...`

For voice call events:

- Point the **Conversation Status Changed** event to `/webhook/call-complete`

For form submissions:

- Create a GHL Funnel with a form and set the webhook URL to `/webhook/form-submit`

Note: Your server must be publicly accessible. Use ngrok for local development:

```
ngrok http 5001
# Copy the https URL and use it as the base for your webhook URLs
```

GHL Custom Fields Expected

The system reads these custom fields from GHL contact payloads:

FIELD KEY	DESCRIPTION
`homeowner_status`	owner / renter / unknown
`monthly_bill`	Monthly electricity bill in AUD
`roof_type`	tile / colorbond / flat / metal
`roof_age`	Roof age in years

HubSpot Setup

HubSpot is supported as a fallback CRM.

Required credentials

```
HUBSPOT_API_KEY=pat-na1-...  # Private app token (not legacy API key)
```

Getting a Private App Token

1. Go to **Settings → Integrations → Private Apps**
2. Click **Create a private app**
3. Under **Scopes**, enable:

- `crm.objects.contacts.read`
- `crm.objects.contacts.write`
- `crm.objects.deals.read`
- `crm.objects.deals.write`
- `conversations.read`
- `timeline` (for notes)

1. Click **Create app** and copy the token

Salesforce Setup

Salesforce is supported as a third fallback.

Required credentials

```
SALESFORCE_USERNAME=your@email.com
SALESFORCE_PASSWORD=yourpassword
SALESFORCE_SECURITY_TOKEN=yourtoken
SALESFORCE_CLIENT_ID=your-connected-app-client-id
SALESFORCE_CLIENT_SECRET=your-connected-app-secret
```

Getting credentials

1. **Security Token:** Go to **Settings → My Personal Information → Reset My Security Token**
2. **Connected App:** Go to **Setup → Apps → App Manager → New Connected App**

- Enable **OAuth Settings**
- Add scopes: `api`, `refresh_token`
- Save and copy the Consumer Key (Client ID) and Consumer Secret

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Feature Matrix

FEATURE	GHL	HUBSPOT	SALESFORCE
Get contact	Yes	Yes	Yes
Create contact	Yes	Yes	Yes
Update contact field	Yes	Yes	Yes
Add tag	Yes	Yes (property)	Yes (field)
Move pipeline stage	Yes	Yes (deal stage)	Yes (opportunity stage)
Create task	Yes	Yes	Yes
Send SMS	Yes	No	No
Add note	No (no native API)	Yes	Yes
Find by phone	Yes	Yes	Yes (SOQL)
Get pipeline stages	Yes	Yes	Yes
Inbound webhooks	Yes (native)	Yes (via subscriptions)	Yes (via outbound messages)

How to Switch CRMs

1. Remove or blank out the current CRM's key in `.env`
2. Add the new CRM's credentials

3. Restart `python main.py`

The `active_crm()` function re-evaluates on every call, so no code changes are needed.

To check which CRM is active:

```
curl http://localhost:5003/api/crm/status
```

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crm_router Public API Reference

All functions are in `integrations/crm_router.py`. All return `None` on failure (never raise).

FUNCTION	PARAMS	RETURNS	GHL	HUBSPOT	SF
`active_crm()`	—	`str` ('ghl'/'hubspot'/'salesforce'/'none')	—	—	—
`all_configured_crms()`	—	`list[str]`	—	—	—
`is_configured()`	—	`bool`	—	—	—
`status()`	—	`{active, ghl, hubspot, salesforce}`	—	—	—
`get_contact(contact_id)`	`str`	contact dict or None	Yes	Yes	Yes
`create_contact(data)`	`dict`	created contact dict or None	Yes	Yes	Yes
`update_contact_field(contact_id, field, value)`	`str, str, any`	dict or None	Yes	Yes	Yes
`add_contact_tag(contact_id, tag)`	`str, str`	dict or None	Yes	Yes	Yes
`move_pipeline_stage(contact_id, stage_id)`	`str, str`	dict or None	Yes	Yes	Yes
`create_task(contact_id, title, due_date)`	`str, str, str`	dict or None	Yes	Yes	Yes
`send_sms(contact_id, message)`	`str, str`	dict or None	Yes	No	No
`add_note(contact_id, note_body)`	`str, str`	dict or None	No	Yes	Yes
`get_pipeline_stages(pipeline_id=None)`	`str \	None`	list	Yes	Yes
`find_contact_by_phone(phone)`	`str`	dict or None	Yes	Yes	Yes

GHL API Client Reference

The underlying `integrations/ghl_client.py` (used directly when not going through crm_router):

FUNCTION	PARAMS	NOTES
`get_contact(contact_id)`	`str`	GET /contacts/{id}
`update_contact_field(contact_id, field, value)`	`str, str, any`	PUT /contacts/{id} with customField
`move_pipeline_stage(contact_id, stage_id)`	`str, str`	POST /contacts/{id}/pipeline-stage
`add_contact_tag(contact_id, tag)`	`str, str`	POST /contacts/{id}/tags
`create_task(contact_id, title, due_date)`	`str, str, str`	POST /tasks/
`send_sms(contact_id, message)`	`str, str`	POST /conversations/messages (type: SMS)
`create_contact(data)`	`dict`	POST /contacts/ (locationId auto-injected)
`get_pipeline_stages(pipeline_id)`	`str`	GET /opportunities/pipelines/{id}
`is_configured()`	—	True if GHL_API_KEY and GHL_LOCATION_ID both set

All functions return `None` on error. Check `config.GHL_API_KEY` before calling if you want to skip gracefully.

Base URL: `https://services.leadconnectorhq.com`

API Version header: `2021-07-28`